Fiscal Year:	FY 2018	**Task Last Updated:**	FY 02/13/2018
PI Name:	Basner, Mathias M.D., Ph.D.		
Project Title:	HERO Twin Astronaut Study Consortium (TASC) Project: Cognition on Monozygotic Twin on Earth		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Philadelphia	**State:**	PA
Zip Code:	19104-4209	**Congressional District:**	2
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2013-14 HERO NNJ13ZSA002N-TWINS
Start Date:	04/14/2014	**End Date:**	04/13/2018
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Charles, John	**Contact Phone:**	281-483-7224
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: Extended to 4/13/2018 per NSSC information (Ed., 1/16/18) NOTE: Extended to 12/31/2017 per PI (Ed., 2/3/17)		
Key Personnel Changes/Previous PI:	February 2018 report: Sarah McGuire, PhD left our working group.		
COI Name (Institution):	Dinges, David Ph.D. (University of Pennsylvania) Gur, Ruben Ph.D. (University of Pennsylvania) Mollicone, Daniel Ph.D. (Pulsar Informatics, Inc.)		
Grant/Contract No.:	NNX14AH27G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This study is part of an integrated compilation of multiple studies led by multiple principal investigators (PIs) at multiple centers which take advantage of a very unique opportunity to look at the effects of space travel on identical twins, one of whom experiences space travel for one year; the other remains Earth-bound for that same year. This study focuses on psychosocial and neurobehavioral experiments to investigate and characterize any differences between the flight subject and the ground subject in cognition, decision making ability, alertness levels, stress, and overall emotional well-being, as a result of the spaceflight environment. We will assess cognitive performance in the twins with the Cognition test battery 3 times pre-, 11 times in-, and 3 times post-flight. Cognition was specifically developed for high-performing astronauts and consists of 10 brief, validated neuropsychological tests that cover a wide range of cognitive domains. The project's specific aims serve to determine how cognitive aspects of the flight subject's phenotype change in response to acute and prolonged exposure to microgravity and other spaceflight stressors, both on a within-subject level (e.g., in-flight vs. pre-flight) and on a between-subjects level (i.e., in comparison to the ground subject's data). These phenotypic data will be shared within and outside the consortium and correlated with findings from the -omics analyses.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The Cognition test battery was specifically designed for the high performing and highly motivated astronaut population. It will thus be a useful tool for the objective assessment of cognitive performance in similar high performing populations on Earth (e.g., medical doctors).
Task Progress:	Cognition is a computerized battery consisting of 10 brief neuropsychological tests that cover a range of cognitive domains and were specifically designed for the high-performing astronaut population. Both Scott Kelly (SK) and Mark Kelly (MK) performed the Cognition battery 4 times pre-flight, 11 times in-flight, and 3 times post-flight. Cognition administration was synchronized with blood draws whenever possible. Cognition data were standardized relative to normative ground data of 15 astronauts (including SK and MK). Overall, the Cognition data do not support a relevant decrease in cognitive performance in-flight by increasing mission duration from 6 to 12 months. There was some indication of a slight decrease in cognitive speed in the 2nd mission half. The only relevant difference between SK and MK was on the Abstract Matching test. Improved accuracy late in-flight and post-flight in MK was not observed in SK. The data suggest that post-flight cognitive performance may be affected by increasing mission duration from 6 to 12 months. There was a slight decrease in cognitive speed and accuracy post-mission. Cognition data were transferred to Dr. Snyder's lab for integration with results from the other TWINS investigators. We have actively been working with the other TWINS investigators on a first publication in the journal Science. We are also working on a companion paper with more detailed Cognition findings.
Bibliography Type:	Description: (Last Updated: 06/17/2026)
Articles in Peer-reviewed Journals	Moore TM, Basner M, Nasrini J, Hermosillo E, Kabadi S, Roalf DR, McGuire S, Ecker AJ, Ruparel K, Port AM, Jackson CT, Dinges DF, Gur RC. "Validation of the Cognition Test Battery for spaceflight in a sample of highly educated adults." Aerosp Med Hum Perform. 2017 Oct 1;88(10):937-46. https://doi.org/10.3357/AMHP.4801.2017 ; PubMed PMID: 28923143 , Oct-2017
Articles in Peer-reviewed Journals	Garrett-Bakelman FE, Darshi M, Green SJ, Gur RC, Lin L, Macias BR, McKenna MJ, Meydan C, Mishra T, Nasrini J, Piening BD, Rizzardi LF, Sharma K, Siamwala JH, Taylor L, Vitaterna MH, Afkarian M, Afshinnekoo E, Ahadi S, Ambati A, Arya M, Bezdan D, Callahan CM, Chen S, Choi AMK, Chlipala GE, Contrepois K, Covington M, Crucian BE, De Vivo I, Dinges DF, Ebert DJ, Feinberg JI, Gandara JA, George KA, Goutsias J, Grills GS, Hargens AR, Heer M, Hillary RP, Hoofnagle AN, Hook VYH, Jenkinson G, Jiang P, Keshavarzian A, Laurie SS, Lee-McMullen B, Lumpkins SB, MacKay M, Maienschein-Cline MG, Melnick AM, Moore TM, Nakahira K, Patel HH, Pietrzyk R, Rao V, Saito R, Salins DN, Schilling JM, Sears DD, Sheridan CK, Stenger MB, Tryggvadottir R, Urban AE, Vaisar T, Van Espen B, Zhang J, Ziegler MG, Zwart SR, Charles JB, Kundrot CE, Scott GBI, Bailey SM, Basner M, Feinberg AP, Lee SMC, Mason CE, Mignot E, Rana BK, Smith SM, Snyder MP, Turek FW. "The NASA Twins Study: A multidimensional analysis of a year-long human spaceflight." Science. 2019 Apr 12;364(6436):eaau8650. https://science.sciencemag.org/content/364/6436/eaau8650.long ; PubMed PMID: 30975860 , Apr-2019
Articles in Peer-reviewed Journals	Lee G, Moore TM, Basner M, Nasrini J, Roalf DR, Ruparel K, Port AM, Dinges DF, Gur RC. "Age, sex, and repeated measures effects on NASA's "Cognition" test battery in STEM educated adults." Aerosp Med Hum Perform. 2020 Jan 1;91(1):18-25. https://doi.org/10.3357/AMHP.5485.2020 ; PubMed PMID: 31852569 , Jan-2020
Articles in Peer-reviewed Journals	Basner M, Moore TM, Hermosillo E, Nasrini J, Dinges DF, Gur RC, Johannes B. "Cognition test battery performance is associated with simulated 6df spacecraft docking performance." Aerosp Med Hum Perform. 2020 Nov;91(11):861-7. https://doi.org/10.3357/AMHP.5602.2020 ; PMID: 33334406; PMCID: PMC7755107 , Nov-2020
Articles in Peer-reviewed Journals	Casario K, Howard K, Cordoza M, Hermosillo E, Ibrahim L, Larson O, Nasrini J, Basner M. "Acceptability of the Cognition Test Battery in astronaut and astronaut-surrogate populations." Acta Astronaut. 2022 Jan;190:14-23. https://doi.org/10.1016/j.actaastro.2021.09.035 ; PubMed PMID: 34803193; PubMed Central PMCID: PMC8601114 , Jan-2022